Lecture: Threats and Threat Actors

Threat Actors / Agents, Attributes of Threat Actors, and Threat Vectors

1. Introduction to Threats in Cyber Security

In cyber security, a **threat** is any **event, action, or circumstance** that has the potential to **harm information systems, networks, or data** by exploiting vulnerabilities.

According to the fundamentals discussed in the uploaded material:

- A **vulnerability** is a weakness in a system
- A **threat** is the potential cause of harm
- An **attack** occurs when a threat is successfully executed

When the **probability of a threat** is combined with the **potential impact**, it is referred to as **risk**.

2. Threat Actors / Threat Agents

A **Threat Actor (or Threat Agent)** is any **individual, group, or entity** that intentionally or unintentionally causes a cyber security incident.

Threat actors can be classified based on **intent, capability, and origin**.

2.1 Insider Threat Actors

Insiders are individuals **within the organization** who have authorized access to systems.

Types of Insider Threats:

- **Malicious insiders** – employees intentionally stealing or damaging data
- **Negligent insiders** – employees who unintentionally cause breaches due to poor security practices
- **Inadvertent insiders** – lack of awareness or training

Examples:

- Employee leaking confidential data
- Weak password usage

- Accidental deletion of critical files

2.2 Outsider Threat Actors

Outsiders operate **outside the organizational boundary**.

Examples:

- Hackers
- Cybercriminals
- Hacktivists
- Nation-state attackers

These attackers usually exploit:

- Network vulnerabilities
- Web application flaws
- Weak authentication mechanisms

2.3 Categories Based on Motivation (from uploaded notes)

a) Political Motivation

- Espionage
- Cyber warfare
- Disrupting national infrastructure
- Making political statements

b) Economic Motivation

- Financial fraud
- Intellectual property theft
- Ransomware attacks
- Credit card fraud

c) Socio-Cultural Motivation

- Ideological goals

- Activism (Hacktivism)
- Curiosity or fame
- Ego gratification

2.4 Other Common Threat Actors

Threat Actor	Description
Script Kiddies	Low-skilled attackers using ready-made tools
Organized Cybercriminals	Highly skilled groups focused on profit
Terrorist Groups	Use cyber attacks to create fear and disruption
Nation-States	Conduct cyber espionage and cyber warfare

3. Attributes of Threat Actors

Attributes of threat actors describe their **capabilities and behavior**, which help security professionals assess risk.

3.1 Motivation

The reason behind an attack:

- Financial gain
- Political influence
- Revenge
- Curiosity
- Sabotage

3.2 Capability (Skill Level)

Threat actors vary in technical expertise:

- Low (script kiddies)
- Medium (organized attackers)
- High (nation-state actors)

Higher capability usually means:

- Sophisticated attacks
- Stealthy techniques
- Use of zero-day vulnerabilities

3.3 Resources

Includes:

- Access to tools and malware
- Infrastructure (botnets, servers)
- Financial backing

Nation-state actors typically possess **high resources**, while individual attackers have limited resources.

3.4 Access Level

- **Insiders** already have system access
- **Outsiders** must first gain unauthorized access

3.5 Intent

Threat actors may be:

- **Intentional** – deliberate attacks
- **Unintentional** – accidental actions (human error)

3.6 Persistence

Some attackers:

- Perform quick attacks
- Maintain long-term presence (Advanced Persistent Threats – APTs)

4. Threat Vectors

A **Threat Vector** is the **path or method used by a threat actor to gain unauthorized access** to a system.

Threat vectors exploit vulnerabilities in:

- Software
- Hardware
- Networks
- Human behavior

4.1 Network-Based Threat Vectors

- Man-in-the-Middle (MITM) attacks
- Denial of Service (DoS / DDoS)
- Packet sniffing
- Session hijacking

4.2 Web-Based Threat Vectors

From the uploaded material:

- SQL Injection
- Code Injection
- File Inclusion attacks
- URL interpretation
- DNS Spoofing

4.3 Malware-Based Threat Vectors

Malware acts as a delivery mechanism for attacks.

Common examples:

- Viruses
- Worms
- Trojans
- Logic bombs
- Backdoors
- Bots

4.4 Email-Based Threat Vectors

- Phishing
- Spear phishing
- Malicious attachments
- Fake links

These attacks primarily exploit the **human layer**, which is highlighted as the weakest link.

4.5 Physical Threat Vectors

- Theft of laptops or mobile devices
- Unauthorized access to hardware
- Hardware tampering
- Counterfeit devices

4.6 Wireless and Mobile Threat Vectors

From your uploaded notes:

- Bluetooth hacking
- Mobile malware

- Overbilling attacks
- Spoofed PDP attacks
- Signaling-level attacks

5. Relationship Between Threat Actors, Attributes, and Threat Vectors

- **Threat actors** decide *who attacks*
- **Attributes** explain *why and how capable they are*
- **Threat vectors** explain *how the attack is carried out*

Effective cyber security requires:

- Identifying threat actors
- Understanding their attributes
- Blocking or mitigating threat vectors